

L11: Why The Optimal Value of q is Equal to Two in a Two-Dimensional World?

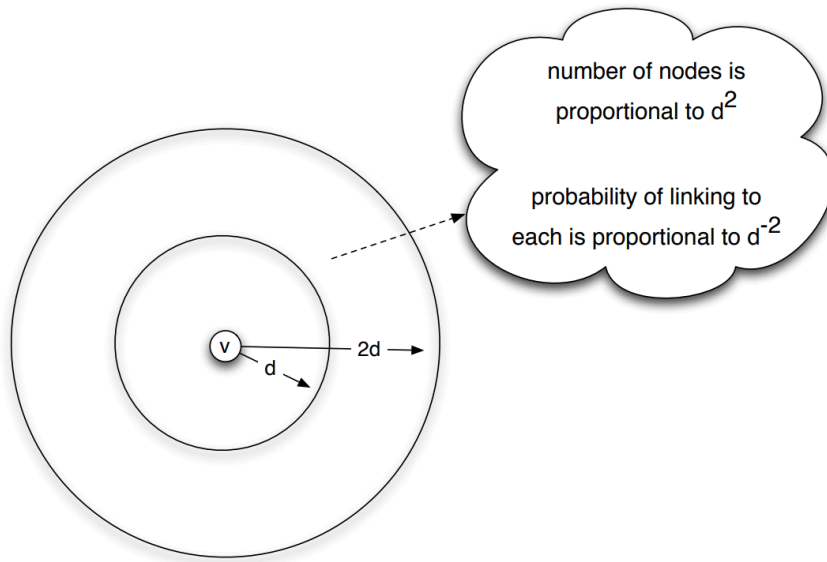


Image Source: Textbook [“Networks, Crowds and Markets”](https://www.cs.cornell.edu/home/kleinber/networks-book) 
(<https://www.cs.cornell.edu/home/kleinber/networks-book>)
by Easley and Kleinberg. Figure 20.7

We can organize distances into different **“scales of resolution”**: something can be around the world, across the country, across the state, across town, or down the block.

Suppose that we focus on a node v in the network – and consider the group of nodes that are at distances between d and $2d$ from v in the network, as shown in the visualization.

If v has a link to a node inside that group, then it can use that link to get closer to the set of nodes that are within that scale of resolution.

So, what is the probability that v has a link to a node inside that group? And how many nodes are expected to be in that distance range?

In a two-dimensional world, *the area around a certain point (node v) grows with the square of the radius of a circle that is centered at v .* So, the total number of nodes in the group that are at a distance between d and $2d$ from v is proportional to d^2 .

On the other hand, the probability that node v links to any node in that group scales as $d(v, w)^{-2}$, when $q = k = 2$. So, if node w is at a distance between d and $2d$ from v , the probability that it is connected to v is proportional to d^{-2} .

Hence, the number of nodes that are at a distance between d and $2d$ from v , and the probability that v connects to one of them, roughly cancel out. This means that the probability that a random edge links to a node in the distance range $[d, 2d]$ is independent of d .

In other words, when $q=k=2$, the long-range links are formed in a way that **“spreads them out”** roughly uniformly over all different resolution scales. This is the key point behind Kleinberg’s result.

If the exponent q was lower than $k=2$, the long-range links would be spread out non-uniformly in favor of larger distances, while if q was larger than $k=2$, the long-range links would be spread out non-uniformly in favor of shorter distances.

Food For Thought

The proof of the previous result, at least for $k=1$ and $k=2$, is clearly written in sections 20.7 and 20.8 of the Easley/Kleinberg textbook. I recommend you go through it.